Language Academy

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UHLB2122 Professional Communication Skills 1

Section 52

Report Writing

Title: Cybersecurity threats are increasing rapidly

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CYBERSECURITY THREATS ARE INCREASING RAPIDLY

1.0 INTRODUCTION

Cybersecurity threats have become increasingly prevalent in today's digital era. According to Kaspersky, Malaysia recorded 19.62 million web-based attacks in the first half of the year of 2024 (Bernama, 2024). Terms such as malware, phishing, and scams are widely known, yet cyber espionage and fraud still affect individuals across all professions. You may have received suspicious calls from unknown numbers, claiming to be officers from government agencies or authorities. These threats are effective because they are packaged in ways that appear authentic, often seem "too real to be fake." However, the use of computers and the internet has become an essential part of modern life, making it impossible to fully isolate ourselves from these risks. Despite the dangers, digital connectivity remains indispensable. This report explores the background, causes, and effects of cybersecurity threats and presents two potential solutions.

2.0 BACKGROUND OF THE PROBLEM

According to Mallick & Nath (2024) and Wielpütz (n.d.), the roots of threats trace back to the earliest days of computing. In 1950, "phone phreaking" emerged, and by the 1960s, the IBM CTSS system revealed one of the first known software vulnerabilities. The first computer virus, Creeper, spread via ARPANET in 1971, followed by Marcus Hess infiltrating over 400 military and research computers in 1986. The 1990s saw the rise of macro viruses, leading to the development of security tools like SSL, while the 2000s and beyond have witnessed a dramatic escalation in malware, phishing, ransomware, and social engineering attacks, growing both in scale and sophistication. As technology has advanced, cyber attackers have shifted their focus not only to exploiting hardware, software, and network vulnerabilities, but also to manipulating human psychology for financial and strategic gain (Mallick & Nath, 2024; Aslan et al., 2023).

2.1 The cause of the problem

The motivations behind cyber-attacks are diverse. Just as criminal acts wouldn't exist in a society where everyone behaves ethically, cybersecurity threats would not arise if all users engaged ethically with digital systems. However, the definition of "ethical" varies from person to person.

According to Turgeman-Goldschmidt (2005), hackers often justify their actions by claiming curiosity, skill demonstration or harmlessness. Some do not view digital offenses as "real crimes" due to the intangible nature of data. They would rationalise their actions using values praised in modern society, such as the pursuit of happiness, curiosity, and knowledge and the demonstration of computer virtuosity, allowing them to present themselves as morally justified or even socially beneficial.

Financial motivation is another dominant factor. According to recent studies, the cyber-attacks result in global economic damages that is expressed in trillions of dollars, and the figure is growing (Aslan et al., 2023). Cybercriminals exploit system vulnerabilities and human gullibility to obtain monetary benefits.

Moreover, the inherent vulnerabilities in computer systems opened the door to exploitation. The digitalisation of daily life and the migration of social activities to the online platform heighten security risks. Software vulnerabilities, network protocol shortcomings, device proliferation, growing attacker expertise and user negligence in internet usage further escalate security concerns in the digital landscape (Mallick & Nath, 2024).

2.2 The effects of the problem

The main effect of cybersecurity is financial loss. Victim suffer direct financial damage through scams or fraud. While organisation bear costs from data breaches, repairs or legal actions. There may also be indirect losses, such as reduced revenue due to piracy or reputational harm. In the first quarter of 2024, police recorded 12,110 online fraud cases in Malaysia, involving losses of RM573.7 million (Mail, 2025).

Other than financial implications, victims may experience psychological impact. Victims may suffer from trauma, stress, anxiety or distrust of digital systems. According to Buse et al. (2023), the investment scam victims tended to feel stupid; the impersonation and identity theft victims typically felt embarrassed; and the bank scam victims typically felt angry. These emotional scars are long-lasting and can affect victims' well-being, family relationships, and self-acceptance.

Cybersecurity attacks can also cause reputational damage. For example, the Facebook data breach raised serious concerns about the platform’s commitment to protecting user data. Many users lost trust in the platform, and even Facebook’s rebranding to Meta failed to fully restore its damaged reputation (Team, 2025).

3.0 POSSIBLE SOLUTIONS

One of the most effective long-term solutions is education. This includes raising public awareness about digital risks, training individuals to recognise and avoid scams, and teaching students the importance of ethical behaviours in digital environment. Digital literacy can be introduced at the school level and reinforced through workplace training. Early education is particularly effective in shaping a strong moral foundation during the critical stages of personality development. The solution can promote a culture of cybersecurity awareness and thus reduces vulnerability to common scams and attacks. Besides, it also encourages responsible behaviour in future technology professionals. However, not all individuals retain or apply the knowledge in real-life situations and sophisticated scams may still deceive even well-educated individuals. Besides, the changing psychological and social influences may still lead some to commit cyber offenses.

Another way is to integrate cybersecurity into system design and not added as an afterthought. Developers must be trained in secure coding practices and encouraged to implement features like encryption, access control and regular patching. From my perspective as a computer science student, I believe future developer like myself should assume responsibility in minimising system vulnerabilities. Thus, the knowledge in cybersecurity should be made mandatory for all developers. By embedding security considerations in the development stage, we can have stronger and more secure systems from the start and reduced the number of exploitable vulnerabilities. However, the advanced technologies evolve quickly, making it difficult to keep up with the rapid changes. Besides, secure development would require more time and cost, and require expertise in the field.

4.0 CONCLUSION

Cybersecurity threats have become an unavoidable part of our digital world. This report has examined the background and evolution of these threats, their causes, impacts and possible solutions. As emphasised throughout, cybersecurity threats affect everyone. Therefore, increasing public vigilance, fostering ethical responsibility, and implementing technical safeguards must be a collective goal. If left unaddressed, cybersecurity threats could undermine public confidence, economic growth, and the integrity of digital infrastructures worldwide. While we may not be able to eliminate cybersecurity threats entirely, we can significantly reduce their impact through education, secure design, and shared responsibility among individuals, developers, organisations, and governments.

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